

RTL Design Sherpa

DDR2/LPDDR2 Characterization Nexys A7 Project Guide 0.90

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1 Document Information

This is the operator / developer guide for the Nexys A7 **DDR2/LPDDR2 Characterization** project (`projects/fpga-systems/NexysA7/mem-ctrl-ip-ddr2/ddr2-characterization`). It explains what the project characterizes, how the harness works, how to build / simulate / program / run it, how the harness CSR is configured, and what the board measured (Chapter 8, added at v0.95).

The memory controller under test — **pumice** — is separate IP (`projects/components/mem-ctrl-ip/research-ip/pumice-ddr2-lpddr2`). This guide treats it as a **black box** driven by the characterization harness; it does not document the controller's internals.

1.1 References

Reference documents

Source	Title
RTL Design Sherpa	`build-perf/host/ADDRESS_MAP.md` (bridge + CSR map)
RTL Design Sherpa	`build-perf/bin/README_a7ddrphy.md` (PHY regen)
RTL Design Sherpa	`flows-litedram-uart/HARNESS_PLAN.md` (baseline flow)
RTL Design Sherpa	`bin/TBClasses/harness/` (shared UART-char collateral)
RTL Design Sherpa	`docs/DDR2_BANDWIDTH_MEASUREMENT.md` (what "bandwidth" means here)
RTL Design Sherpa	`build-perf/reports/` (the raw records behind Chapter 8)
Micron	`MT47H64M16HR-25E` DDR2 SDRAM datasheet
ARM	AMBA AXI4 / AXI4-Lite / APB Protocol Specifications

1.2 Terminology

DFI

DDR PHY Interface. The command/data protocol between the memory controller and the PHY. Here pumice emits a phase-packed flat DFI v2.1 bus.

a7ddrphy

LiteDRAM's Artix-7 DDR2 PHY (4:1 serdes, 800 Mbps). Uses Xilinx OSERDESE2/ISERDESE2/IDELAYE2 and therefore does **not** simulate in Verilator.

GEAR / DFI_RATE

The DFI phase count. `GEAR_RATIO = log2(DFI_RATE)`. Must match between the RTL build and the sim model or reads corrupt.

Engine

A master-side AXI4 traffic generator: the write engine emits an LFSR data pattern; the read engine re-reads and CRC-checks it.

Leveling

Firmware-driven a7ddrphy read/write training over the PHY-CSR passthrough window; there is no hardware leveling FSM.

1.3 Revision History

Revision history

Version	Date	Notes
0.90	2026-07-18	Initial project guide

2 Overview

2.1 What it characterizes

This project performs end-to-end, board-level **performance, latency, and data-integrity characterization** of a DDR2/LPDDR2 memory controller on real silicon. It drives the onboard Micron `MT47H64M16HR-25E` DDR2 (x16, single-rank, 128 MiB, up to 800 Mbps) through the controller against the real device, and reports throughput (MB/s), latency histograms, and per-workload CRC integrity across burst-length / stride / gap / page-policy / scheme sweeps.

There is no soft CPU in the FPGA — an off-board Python **host** drives the harness over the FTDI USB-UART (115200 8N1). The same host programs run against a cocotb simulation (Chapter 6).

2.2 The three flows

The project is organized as sibling "flows" plus a shared framework:

The characterization flows

Flow	Controller (DUT)	What it is
`build-perf/`	pumice DDR2 (black box) + DFI + a7ddrphy	**Primary** flow: UART-driven host + Vivado build of the pumice harness on the Nexys A7.
`flows-litedram-uart/`	LiteDRAM `litedram_core`	Apples-to-apples baseline: LiteDRAM's own DDR2 controller (self-initializing, own PLL/PHY) driven by the **same** engines/taps/host.
`ddr2_char_framework/`	—	Shared instrumentation: harness RTL, the engine spine (`char_engine_block`) and its pumice wrapper

Flow	Controller (DUT)	What it is
		(ddr2_char_macro), the 1→6 AXIL bridge, and the cocotb DV.

The naming convention is `flows-<controller>-uart/` for on-Nexys builds (host over UART, no CPU). pumice is separate IP under `projects/components/mem-ctrl-ip/research-ip/pumice-ddr2-lpddr2/`.

2.3 Board and tools

Board and tool requirements

Item	Detail
Board	Digilent Nexys A7-100T, Artix-7 `xc7a100tcsg324-1`
DRAM	Onboard Micron `MT47H64M16` DDR2 x16, `ROW_WIDTH = 13`
Build	Vivado (batch mode)
Host	Python 3 + `pyserial`
Sim	Verilator + cocotb / cocotb_test / pytest
PHY regen (only)	Python 3.10 + migen / litex 2024.12 / litedram 2024.12

2.4 Workload families

The characterization sweeps four access-pattern families (`build-perf/host/pumice_char.py`):

Access-pattern families

Family	Access	Purpose
`incremental`	contiguous linear	best-case sequential
`row_major`	bounded to one page	guaranteed page HIT
`col_major`	stride = one row, same bank	guaranteed page MISS (worst case)
`col_major_interleaved`	stride = one bank	activates the bank pipeline

Controller-config presets (`baseline`, `bank_interleave`, `open_page`, `inorder`, `reorder`, `happy_hybrid`, `fast_refresh`, `slow_refresh`) each flip one lever from baseline; they are

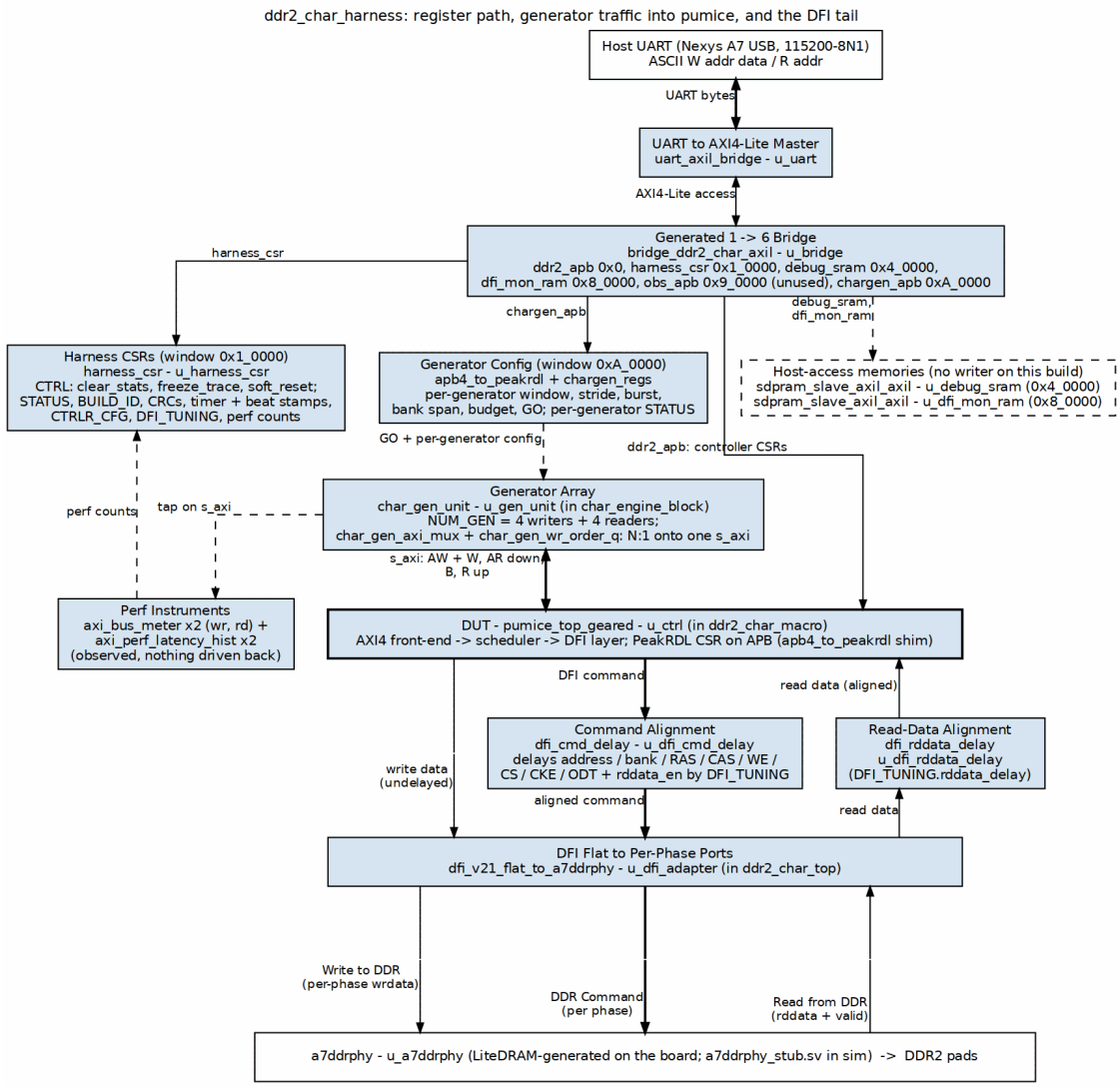
driven over the pumice APB slave (black box). Run profiles `smoke` / `matrix` / `full` are shared by sim and board.

3 How It Works

3.1 The transport spine

Host commands reach the DUT over one UART link that fans out to four AXI4-Lite slaves:

3.1.1 Figure 2.1: DDR2 Characterization Harness Spine



DDR2 characterization harness spine

Source: 01_harness_spine.dot

The UART wire protocol is ASCII, line-based, 115200 8N1: a write is `W <hex-addr32> <hex-data32>\n` → the FPGA replies `OK\n`; a read is `R <hex-addr32>\n` → replies `0x<hex-data32>\n`. The bridge RTL is

projects/components/utility-ip/converters/rtl/uart_to_axil4/uart_axil_bridge.sv; the host framing is UARTAxiBridge (projects/components/utility-ip/converters/bin/uart_axi_bridge.py).

3.2 The 1→6 AXIL bridge

bridge_ddr2_char_axil is generated from ddr2_char_framework/rtl/bridges/configs/bridge_ddr2_char_axil.toml (one master host, six slaves: ddr2_apb, harness_csr, debug_sram, dfi_mon_ram, the unused obs_apb expansion slot, and chargen_apb). It decodes those windows and routes each host transaction to the right slave via per-slave adapters.

3.3 Key RTL blocks

Key RTL blocks

File	Role
build-perf/rtl/ddr2_char_top.sv	FPGA pin-level top: wraps the harness, the flat-DFI→per-phase adapter, and the a7ddrphy black box; MMCM clock synthesis (sys / sys2x / sys4x / sys4x_dqs), IDELAYCTRL, DDR2 pads.
build-perf/rtl/ddr2_char_harness.sv	Internal integration: uart_axil_bridge + bridge_ddr2_char_axil + harness_csr + debug_sram + dfi_mon_ram + ddr2_char_macro + char timer + LED / 7-seg.
projects/fpga-systems/rtl/mem_char_framework/rtl/char_engine_block.sv	The DUT-agnostic spine: chargen_regs (per-generator config behind the chargen_apb window), the generator arrays (axi4_master_wr_injector / axi4_master_rd_crc_check), the N:1 merge (char_gen_axi_mux + char_gen_wr_order_q, inside char_gen_unit) onto one AXI4 master, and the perf taps (bus meters + latency histograms) on that port. Shared by the pumice and LiteDRAM flows.
ddr2_char_framework/rtl/	Wraps char_engine_block around

File	Role
ddr2_char_macro.sv	pumice: the block's AXI4 master drives the controller's s_axi, and the APB CSR / DFI / runtime controls pass through.
projects/fpga-systems/rtl/mem_char_framework/rtl/harness_csr.sv	The AXIL CSR slave (Chapter 5). Hand-written (self-clearing pulses, latches, PHY passthrough).
ddr2_char_framework/rtl/dfi_v21_flat_to_a7ddrphy.sv	Combinational adapter: pumice's phase-packed flat DFI v2.1 → a7ddrphy per-phase ports (NPHASES = 4).
ddr2_char_framework/rtl/a7ddrphy_stub.sv	Port-shape black box of a7ddrphy for Verilator (real body swapped in by Vivado).

3.4 The characterization engines

Inside char_engine_block (instantiated by ddr2_char_macro, and by the LiteDRAM flow's char_engine_harness), master-side AXI4 engines generate and check traffic against the controller's s_axi port:

- **Write engine** (axi4_master_wr_injector) — emits an LFSR data pattern across the programmed address pattern, and latches an expected CRC.
- **Read engine** (axi4_master_rd_crc_check) — re-reads the same pattern and computes an actual CRC, plus a per-beat mismatch count.

Alongside them, **perf taps** — axi_bus_meter (write and read) and axi_perf_latency_hist — observe the internal AXI wires to produce the producer/backpressure/starve/idle buckets and the latency histogram read back in Chapter 5.

3.5 The controller, DFI, and PHY (black boxes)

The controller (**pumice**) sits inside ddr2_char_macro between the AXI engines and the flat DFI output. Its configuration registers are reached over the ddr2_apb bridge slave at base 0x0. This guide documents only that the harness drives it — not its internals.

pumice's flat DFI v2.1 bus is adapted to the a7ddrphy per-phase interface, then to the DDR2 pads. **a7ddrphy does not simulate in Verilator** (it uses Xilinx SERDES/IDELAY primitives); the sim connects at the DFI level with a behavioral model instead (Chapter 6). On hardware, a7ddrphy calibration is driven by firmware (the host) over the PHY-CSR passthrough window — there is no hardware leveling FSM.

4 Test Methodology

This chapter describes what is exercised on the FPGA block (Figure 2.1) and how throughput, latency, and data integrity are measured.

4.1 What is under test

The **pumice** memory controller (black box) driving the real onboard DDR2, stimulated by the two master-side AXI4 engines. Every measurement is taken on the internal AXI wires between the engines and the controller, so the numbers reflect the controller + PHY + device as a whole.

4.2 What is measured, and how

What is measured

Quantity	How it is obtained
Throughput	`beats × bytes_per_beat / (TIMER_CYCLES × 10 ns)` — the timer bounds the run in `sys_clk` cycles
Latency histogram	`OBS_HIST_SEL` selects bus (rd/wr) + metric (AR→firstR / AR→RLAST) + bin; `OBS_HIST_COUNT` / `OBS_HIST_TOTAL` read it back. Bin *b* covers [2^b, 2^(b+1)) cycles
Data integrity	`CRC_ACTUAL` vs `CRC_EXPECTED` (`CRC_MATCH`), plus `BEATS_MISM`
Utilization	`OBS_{RD,WR}_{PROD,BP,STARV,IDLE}` bus-meter buckets

A measurement run: program the engine config (address pattern, burst length, txn count, LFSR seed), `clear_stats`, kick with `CTRL.start_wr` / `start_rd`, poll `STATUS.wr_done` / `rd_done`, then read the timer, CRC, histogram, and meters. The write engine latches the expected CRC as it generates the pattern; the read engine re-reads and computes the actual CRC.

4.3 Workloads

Characterization workloads

Axis	Values
Access pattern	`incremental`, `row_major`, `col_major`, `col_major_interleaved`
Controller preset	`baseline`, `bank_interleave`, `open_page`, `inorder`, `reorder`, … (over the pumice APB slave)

Axis	Values
Burst / count / stride	swept per `--char-profile` (smoke / matrix / full)

4.4 The oracle

Integrity is judged against a deterministic **LFSR pattern** + **CRC-32**: the write engine's `CRC_EXPECTED` must equal the read engine's `CRC_ACTUAL` with `BEATS_MISM = 0`. The same golden logic runs in sim (Chapter 6).

4.5 Measurement pitfalls

The full treatment — the four candidate denominators, how to choose the measurement window, the bounds a number should be checked against, and the complete pitfall list — is `docs/DDR2_BANDWIDTH_MEASUREMENT.md` in this area. Read it before publishing a bandwidth figure. What follows is only the operational short list.

- `STATUS.any_error` is **sticky** — clear it with `clear_stats` between phases or every later run looks wedged.
- **The mismatch counter is PER READER.** `beats_mismatched()` with no argument checks reader 0 only; on a 4+4 run that leaves three readers unverified and reports them clean. Sum across the active engines.
- **Sweep points are not independent unless you re-fill.** A point that leaves wrong data in cells is inherited by every later point that reads the region, and the inherited damage is reported as the later point's own result. This made the first 192-point gap sweep unreadable (pumice BUG-014 (was PUMICE-037)).
- **A constant repeated across runs is stale state, not a race.** A race varies. An identical mismatch count at eight consecutive gaps means one artifact is being re-read eight times.
- **Writers and readers on disjoint banks means nobody verifies the writers' banks.** A defect can sit there for a whole run and surface only when a later configuration happens to put a reader on a bank earlier points wrote — which is exactly how pumice BUG-014 (was PUMICE-037) appeared, in the last block of a 192-point sweep.
- Keep board runs `txn ≤ 1024` (read-engine CAM ceiling) for clean completion; the *rate* stays valid at scale.
- `col_major` stride × large txn can alias past the 128 MB device — bound it.

- The sim must be built at the board's exact `DFI_RATE` / GEAR or it passes while the board fails (Chapter 6).

5 Build and Run

All commands assume the Python environment is sourced (sets `REPO_ROOT`, `SIM=verilator`, `PATH`, `PYTHONPATH`):

```
cd /path/to/RTLDesignSherpa && source env_python
cd projects/fpga-systems/NexysA7/mem-ctrl-ip-ddr2/build-perf
```

5.1 `build-perf` make targets

`UART ?= /dev/ttyUSB1`, `BAUD ?= 115200`, `VIVADO ?= vivado`.

build-perf make targets

Target	Action
`make lint`	Verilator lint of `ddr2_char_top` via the harness filelist
`make project` / `make synth`	create Vivado project / synth-only + reports
`make bitstream`	full synth + impl + bitgen (~10–30 min) → `bitstream/ddr2_char.bit`
`make bitstream-ila`	same, plus an ILA on the DFI boundary → `.bit` + `.ltx`
`make program`	flash the board
`make utilization` / `make timing`	print the latest reports
`make smoke`	`host/run_smoke.py` — link + one linear WR/RD integrity pass
`make status`	`host/ddr2_char.py` — one-shot BUILD_ID + status dump
`make level`	`host/pumice_master.py --level-only` — a7ddrphy leveling
`make simple`	`host/pumice_master.py --simple` — init + one write→read pass
`make characterize`	`host/pumice_master.py --full` — full workload sweep
`make sweep-rddly` / `make train-deskew`	PHY timing sweeps
`make host-test`	`pytest host/test_pumice_master.py`

Target	Action
	(mock UART, no board)
`make sim` / `sim-smoke` / `sim-simple` / `sim-level`	run the **same** host programs in cocotb sim
`make clean` / `clean-all`	remove build artifacts

The other Makefiles: `flows-litedram-uart/Makefile` (`make regen` / `make bitstream` / `make program` for the LiteDRAM baseline) and `ddr2_char_framework/dv/tests/Makefile` (the cocotb macro suite: `make run`, `run-smoke`, `run-<shape>`, and parallel `run-{gate,func,full}-parallel`).

5.2 The host programs (`build-perf/host/`)

The port defaults to `--port auto`, which probes every `/dev/ttyUSB*` and keeps the board that answers `BUILD_ID == 0x44445232` ("DDR2").

`run_smoke.py` — the first thing to run after flashing. Programs a linear WR+RD workload, kicks both engines, and checks CRC / cycles / perf.

```
python3 host/run_smoke.py --port /dev/ttyUSB1 --txn 1024 --blen 8 --
seed 0xDEADBEEF
```

Pass = BUILD_ID matches, both engines done, `CRC_ACTUAL == CRC_EXPECTED`, `BEATS_MISMATCHED == 0`, and `TIMER.pass`.

`pumice_master.py` — the orchestration program; one mode is required:

```
python3 host/pumice_master.py --port /dev/ttyUSB1 --level-only   # PHY
read/write leveling
python3 host/pumice_master.py --port /dev/ttyUSB1 --simple       #
init + one write→read pass
python3 host/pumice_master.py --port /dev/ttyUSB1 --full         #
full workload sweep
python3 host/pumice_master.py --port /dev/ttyUSB1 --char --char-
profile matrix --char-scale 1000 --csv out.csv
```

Useful options: `--no-level`, `--rd-phase`, `--rd-delay`, `--char-configs`, `--char-scale N` (≈1000 on FPGA), `--level-cache JSON`, `--clk-mhz`.

`ddr2_char.py` — the driver library plus a one-shot status CLI (`python3 host/ddr2_char.py --port auto`). It exports `DDR2CharDriver`, the base-address constants, and the enum constants used throughout.

5.3 The performance sweeps (`build-perf/bin/`)

Three scripts, each isolating one axis. Run them from `build-perf/` after `make program`; all three write JSON or a table and none need arguments.

`axlen_sweep.py` — bandwidth against AXI burst length at a fixed outstanding budget. This is the headline read/write figure and the Little's-law fit.

`outstanding_sweep.py` — bandwidth against the number of bursts in flight, at a fixed AxLEN. This is the DIRECT test of the latency-bound claim: if a shortfall is Little's law, the knee sits near `latency/AxLEN` and moves as `1/AxLEN`. A knee well before that names whatever the real limit is instead. The runtime `max_outstanding` dial and the 32-deep ceiling exist for this sweep.

`bank_gap_sweep.py` (plot with `plot_bank_gap.py`) — N writers and N readers running CONCURRENTLY, one bank apiece, across inter-burst gap 0..15, three address orders, at 4+4 / 3+3 / 2+2 / 1+1 engines. It answers how much idle the controller absorbs before bandwidth falls.

```
python3 bin/axlen_sweep.py
python3 bin/outstanding_sweep.py
python3 bin/bank_gap_sweep.py && python3 bin/plot_bank_gap.py
reports/bank_gap_sweep.json
```

Useful env: `GENS`, `GAPS` (a named sweep — `full`, `knee`, `ends` — or a literal list), `TXN`, and `PREFILL`. **Leave `PREFILL` at its default `point`.** It re-fills the whole device before every measurement so points are independent; `PREFILL=once` exists only for a deliberate damage-accumulation experiment, not to go faster. The gap range stops at 15 because the CSR field is four bits — asking for 16 programs 0 and silently re-measures back-to-back under a different label.

Reading the output: the knee column is the largest gap still within 3% of the gap-0 value. A FLAT curve at high generator counts is a result, not a broken axis — a gap only bends the curve once aggregate demand drops below what the controller can deliver, and four generators at gap 15 are still over-subscribing it. Points whose data failed verification are ringed in the plots and listed as `FAIL:` lines with their stray-beat count; their bandwidth is still a real measurement, but the configuration is not a clean operating point.

5.4 Typical bring-up sequence

```
make bitstream
make program
make smoke   UART=/dev/ttyUSB1      # link + integrity
make level                          # find the read/write eye
make simple                         # one clean write→read
make characterize                   # full sweep -> CSV
# no board? prove the same programs in sim:
make sim
```

6 Harness CSR Configuration

6.1 Bridge address map

The UART bridge fans out to four AXI4-Lite slaves. Bases come from `bridge_ddr2_char_axil.toml` (mirrored in `build-perf/host/ADDRESS_MAP.md` and as constants in `ddr2_char.py`):

Bridge slave address map

Constant	Slave	Base	Size	Protocol
DDR2_APB_BASE	ddr2_apb	0x0000_0000	4 KB	APB — pumice controller CSR (black box)
HARNESS_CSR_BASE	harness_csr	0x0001_0000	4 KB	AXIL — this chapter
DEBUG_SRAM_BASE	debug_sram	0x0004_0000	256 KB	AXIL 64b — MonBus/DFI trace ring
DFI_MON_RAM_BASE	dfi_mon_ram	0x0008_0000	4 KB	AXIL — DFI cmd observability

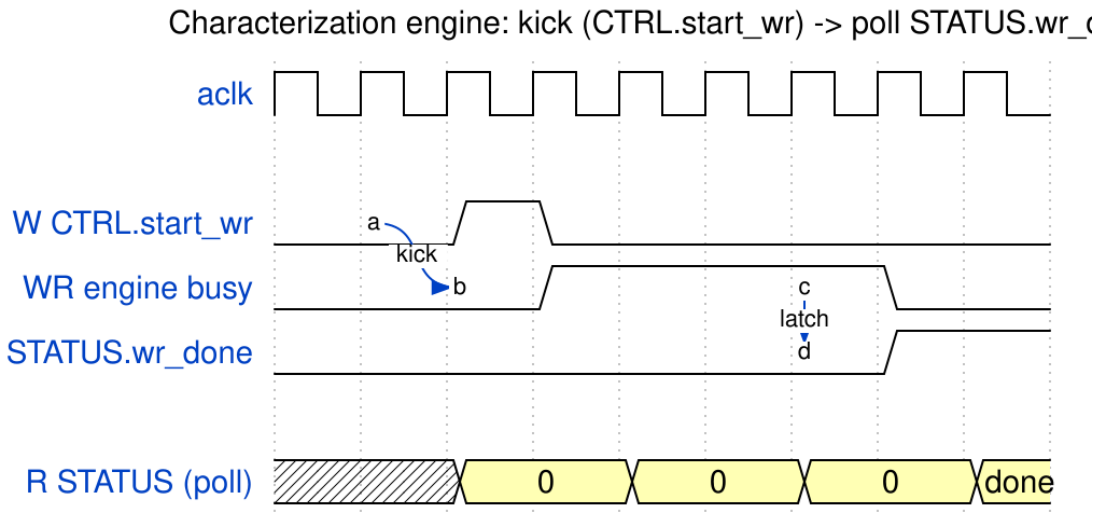
6.2 By-name access (RDL → regmap)

`harness_csr.sv` is hand-written; `harness_csr.rdl` is a descriptor of it, from which the by-name host regmap is generated. Registers are accessed by name via `UartRegisterMap`, never by hardcoded offset. Regenerate after editing either:

```
python3 bin/peakrdl_generate.py \
  projects/fpga-systems/rtl/mem_char_framework/rtl/harness_csr.rdl \
  --regmap --docs-only --no-html --no-markdown \
  --regmap-output
projects/fpga-systems/rtl/mem_char_framework/dv/tbclasses/harness_csr_
regmap.py
```

A consistency test (`build-perf/host/test_harness_regmap_consistency.py`) asserts the generated regmap matches the SV header table.

6.2.1 Waveform 5.1: Engine Kick and Poll



Engine kick and poll

Source: 01_engine_kick.json

6.3 Control / status (offsets relative to HARNESS_CSR_BASE)

Control / status registers

Register	Offset	Field	Bits	Access	Description
CTRL	0x00	start_wr	[0]	W	Pulse: kick the write engine
CTRL	0x00	start_rd	[1]	W	Pulse: kick the read engine
CTRL	0x00	clear_stats	[2]	W	Pulse: clear perf + sticky errors
CTRL	0x00	freeze_trace	[3]	RW	Latch: freeze perf counters
CTRL	0x00	soft_reset	[4]	W	Pulse: soft reset (does **not** clear sticky errors)
STATUS	0x04	wr_done / rd_done	[0] / [1]	R	Engine completion
STATUS	0x04	wr_erro	[2] / [3]	R	Per-engine error

Register	Offset	Field	Bits	Access	Description
		r / rd_error			
STATUS	0x04	any_error	[4]	R	Sticky OR of errors — cleared by clear_stats
STATUS	0x04	init_done / init_fail	[6] / [7]	R	Controller init status
DBG_WR_PTR	0x08	value	[31:0]	R	Words written to debug_sram since clear
DBG_OVERFLOW	0x0C	overflow	[0]	R	Sticky trace-overflow
CRC_EXPECTED	0x10	value	[31:0]	R	Write-engine expected CRC
CRC_ACTUAL	0x14	value	[31:0]	R	Read-engine actual CRC
CRC_MATCH	0x18	match	[0]	R	CRC_ACTUAL == CRC_EXPECTED
CRC_MATCH	0x18	exp_valid / act_valid	[1] / [2]	R	CRC captured
CRC_MATCH	0x18	beats_mism_nz	[3]	R	Any beat mismatched
SCRATCH	0x1C	value	[31:0]	RW	Host link ping
BUILD_ID	0x20	value	[31:0]	R	0x44445232 (“DDR2”)
BEATS_MISM	0x24	value	[31:0]	R	Beats mismatched

6.4 Timer

Characterization timer registers

Register	Offset	Bits	Access	Description
TIMER_CTRL	0x28	[0] clear	W	Pulse: clear the cycle counter
TIMER_STATUS	0x2C	[0] done, [1] running, [2] pass	R	Timer state
TIMER_CYCLES_LO/HI	0x30 / 0x34	[31:0]	R	64-bit cycle count (10 ns/cycle)
TIMER_EXP_BEATS	0x38	[31:0]	RW	Beat-count stop trigger (0 = disable)
TIMER_{R,W}_{FIRST,LAST}_{LO,HI}	0x40–0x5C	[31:0]	R	First/last R/W beat timestamps (64-bit each)

6.5 Runtime controller cfg + DFI tuning

Runtime controller config and live DFI tuning

Register	Offset	Field	Bits	Access	Description
CTRLR_CFG	0x60	memtype	[0]	RW	0 = DDR2, 1 = LPDDR2
CTRLR_CFG	0x60	t_phy_wrlat	[15:8]	RW	PHY write latency (see Chapter 7 — board wants **0**)
CTRLR_CFG	0x60	t_rddata_en	[23:16]	RW	Read-data-enable latency
CTRLR_CFG	0x60	rd_in_order	[24]	RW	Force in-order reads
CTRLR_CAP	0x64	cap_lookahead_max	[3:0]	RW	Advertised OOO look-ahead depth
CTRLR_CAP	0x64	cap_synth_mask	[7:4]	RW	Advertised feature mask

Register	Offset	Field	Bits	Access	Description
DFI_TUNING	0x68	cmd_delay	[3:0]	RW	Live DFI cmd delay (reset 1) — no rebuild
DFI_TUNING	0x68	rddata_delay	[7:4]	RW	Live DFI read-data delay (reset 0)

6.6 a7ddrphy calibration passthrough (hardware only)

Indirect access to the a7ddrphy's read/write-leveling knobs; firmware-driven, no HW FSM. Meaningless in sim (the stub ignores it). Knob map: `rtl-vivado/a7ddrphy/a7ddrphy_csr_map.txt`.

a7ddrphy calibration passthrough

Register	Offset	Bits	Access	Description
PHY_CSR_ADDR	0x80	[9:0]	RW	a7ddrphy CSR word index
PHY_CSR_WDATA	0x84	[31:0]	RW	Value to write
PHY_CSR_CTRL	0x88	[0] pulse	W	Drive one CSR-bus write
PHY_CSR_RDATA	0x8C	[31:0]	R	dat_r for the current PHY_CSR_ADDR

6.7 Engine config (WR 0x100–0x128, RD 0x180–0x1A8)

The read block mirrors the write block at +0x80.

Write-engine config (read engine identical at +0x80)

Register	Offset	Field	Bits	Description
WR_START_ADDR	0x100	value	[31:0]	Start address
WR_STRIDE_0/1	0x104 / 0x108	value	[23:0]	Signed strides
WR_WRAP_MASK_0/1	0x10C / 0x110	value	[31:0]	Address wrap masks
WR_BLEN_TXN	0x114	burst_len	[7:0]	AXI burst length
WR_BLEN_TXN	0x114	txn_count	[23:8]	Transaction count
WR_BLEN_TXN	0x114	gap	[27:24]	Inter-burst gap

Register	Offset	Field	Bits	Description
WR_AXI_ATTR	0x118	axi_id	[7:0]	AXI ID
WR_AXI_ATTR	0x118	id_mode	[9:8]	0 fixed / 1 counter / 2 LFSR
WR_AXI_ATTR	0x118	axi_size	[12:10]	AXI size (AXI_SIZE_8 = 3)
WR_AXI_ATTR	0x118	axi_burst	[14:13]	0 fixed / 1 incr / 2 wrap
WR_AXI_ATTR	0x118	data_mode	[15]	Data pattern mode
WR_LFSR_SEED	0x11C	value	[31:0]	Pattern seed (must match RD for CRC)
WR_HASH_SEED0/1/2	0x120–0x128	value	[31:0]	Hash seeds

6.8 Perf observability (0x1C0–0x1E8)

All 32-bit; cleared by CTRL.clear_stats, frozen by CTRL.freeze_trace.

Perf observability (bin b covers [2^b, 2^(b+1)) cycles)

Register	Offset	Field	Bits	Description
OBS_RD_{PROD,BP,STARV,IDLE}	0x1C0–0x1CC	value	[31:0]	Read data-channel meter buckets
OBS_WR_{PROD,BP,STARV,IDLE}	0x1D0–0x1DC	value	[31:0]	Write meter buckets
OBS_HIST_SEL	0x1E0	bus	[0]	0 = read, 1 = write
OBS_HIST_SEL	0x1E0	metric	[1]	0 = AR→firstR / AW→B, 1 = AR→RLAST
OBS_HIST_SEL	0x1E0	bin	[5:2]	Histogram bin index (0–15)
OBS_HIST_COUNT	0x1E4	value	[31:0]	Count in the selected bin
OBS_HIST_TOTAL	0x1E8	value	[31:0]	Total txns on the selected metric

7 Simulation / Silicon Equivalence

This project follows the shared UART-characterization methodology (`bin/TBClasses/harness/`): **one authored-once host program runs byte-for-byte identically on the FPGA and in a cocotb simulation.** The equivalence boundary is the ASCII `W/R` byte stream on the UART wire; only the bottom `ByteChannel` differs — `SerialChannel` (pyserial) on the board vs `CocotbUartChannel` on the DUT pins in sim.

7.1 Testbenches

Simulation testbenches

tb_top	Wraps	Drives cfg via
`ddr2_char_uart_tb_top.sv`	the **full** `ddr2_char_harness` (real UART bridge + harness_csr + 1→N bridge + engines + pumice)	the real UART pins (cocotb `UARTMaster/Monitor`)
`ddr2_char_macro_tb_top.sv`	the inner `ddr2_char_macro`	direct ports + an APB BFM

The UART tb is the equivalence vehicle: it exposes real UART pins so the sim consumes the identical byte stream the host sends to silicon. The DFI side is wired to internal `phy_dfi_*` nets so the framework's `DFISlavePHY` + `MemoryModel` loopback auto-binds by prefix. **There is no a7ddrphy in sim** — it reproduces the digital DFI handshake, not the analog eye.

7.2 How the same program runs in sim

`test_ddr2_char_uart.py` builds the `DFISlavePHY` + `MemoryModel` backend, then:

```
r = await cocotb.external(prog)()   # prog is an UNMODIFIED pumice_master routine
```

`make_uart_channel(dut, clock, clks_per_bit)` returns a `CocotbUartChannel` that drives the UART pins; the synchronous host program runs in a worker thread under `cocotb.external`, and the channel's read/write are `cocotb.function` wrappers that step the sim. The baud is lowered for sim (`UART_BAUD = 6_250_000` → `CLKS_PER_BIT = 16` vs 868 on board) — bit-timing only, the byte stream is unchanged. `make sim` / `sim-smoke` / `sim-simple` / `sim-level` run the same programs the board runs.

7.3 Match the board's exact DFI config

`GEAR_RATIO = log2(DFI_RATE)` and **must** match the compile-time `DFI_RATE` or reads corrupt. The test threads `TEST_DFI_RATE` / `TEST_GEAR_RATIO` / `TEST_DRAM_BEAT_BYTES` into both the SV `parameters={...}` and the host via `drv.set_dfi_phase(...)`. The board is built at `DFI_RATE = 4` (GEAR = 2, AXI 64 / DRAM beat 32); a rate-2 / GEAR-1 sim will **pass while the board fails** — it skips the very paths that break. Named variants (`..._smoke_rate4`, `..._smoke_rate4_x16`, `..._pagehit_rate4_x16`, ...) exist to run the board's real geometry.

7.4 What the sim can and cannot reproduce

- **CAN:** everything digital — the controller/engines, the bridge RTL, DFI-level handshakes, register logic. This is where it earns its keep.
- **CANNOT:** analog PHY behavior. Xilinx SERDES/IDELAY in a7ddrphy do not simulate, so read-eye / DQS-gate / pin-LOC problems are silicon-only.

8 Troubleshooting

8.1 Current board state

Sim timing is closed; on-board DDR2 runs but is **gated by PHY calibration**. A clean read-leveling eye is found only at **`t_phy_wrlat = 0`** (bitslip 0, taps 0–11, centre tap 5). The default of `t_phy_wrlat = 4` used by `--simple` / `--char` breaks the write path ("no passing tap").

Known-good board config: `t_phy_wrlat = 0`, `t_rddata_en = 6`, `rd_phase = 0`, `rddata_delay = 8`, `cmd_delay = 1`.

After the runtime page-policy fix, OPEN vs CLOSE page policy moved read throughput from ~12.7 MB/s to ~112 MB/s (8.8×) and write to ~44 MB/s (3.5×) (measured 2026-07, pre-sdpram-fix). For reference, LiteDRAM on the same board reaches ~600 MB/s peak (300 MT/s memtest).

8.2 Sticky errors look like a wedge

`STATUS.any_error` is **sticky** and is cleared by `CTRL.clear_stats`, **not** `soft_reset`. A single failed read latches `rd_error`; every subsequent engine wait then returns False and looks like a hardware wedge that isn't one. Clear stats between phases. Note `pumice_master.wait_engine()` waits on **one** engine (write-then-read is phased); a "wait for both" would hang.

8.3 Known bugs / bounds

Known issues and bounds

Symptom	Cause / workaround
Read engine wedges at ~4790 txns	`rd_cmd_cam` ceiling — keep board runs `txn ≤ 1024` for clean completion; the perf *rate* is still valid at scale
`row_major` wedges immediately	every burst is a row-miss — needs the wedge fix before it can be characterized
Intermittent 1–11 beat mismatches at scale	clean at `txn ≤ 1024`
`col_major` stride overflow	stride 16 KB × txn > 8192 exceeds 128 MB → address alias; host bounds `col_major` to `txn ≤ 8192`
`synth_mask_obs` / `lookahead_max_obs` read 0	init never set `CTRLR_CAP` — advertise capability first

8.4 Cannot find the board

The USB-UART re-enumerates across reboots/replugs. `--port auto` (default) probes each `/dev/ttyUSB*`, reads `BUILD_ID`, and picks the one answering `0x44445232`. If none respond it raises "no pumice DDR2 char harness responded … is the board powered and programmed?" — confirm the bitstream is flashed and the FTDI cable is free.

8.5 a7ddrphy is not simulatable

Verilator cannot model OSERDESE2 / ISERDESE2 / IDELAYE2, so sim connects at the DFI level with `DFISlavePHY` + `MemoryModel`. Read-eye / DQS-gate / pin-LOC problems are silicon-only. Regenerating the real PHY requires **Python 3.10 only** (migen's CSR-name tracer breaks on 3.11/3.12):

```
uv venv --python 3.10 /tmp/litex-venv310 && source /tmp/litex-
venv310/bin/activate
uv pip install migen "litex==2024.12" "litedram==2024.12" pyyaml
cd projects/fpga-systems/NexysA7/mem-ctrl-ip-ddr2/build-perf
python3 bin/elaborate_a7ddrphy.py --out
rtl-vivado/a7ddrphy/a7ddrphy_generated.v
python3 bin/elaborate_a7ddrphy.py --dump-csr-map      # CSR offsets
for firmware
```

8.6 debug_sram trace ring is disabled on this build

`DEBUG_SRAM_WORDS` was shrunk 32768 → **512** — the full ring needed ~44 K distributed-RAM cells (2.4× over the 100T budget) and blocked `place_design`. The 256-KB address *window* is unchanged (accesses > 2 KB alias into the ring); the ring is effectively unused here. Raise it on a larger device.

8.7 LiteDRAM baseline flow

`flows-litedram-uart` is wired and lint-clean but board bring-up is gated. The default `./regen.sh` builds an empty BIOS ROM (`--no-compile-software`) so the core never asserts `init_done` — use `./regen.sh --bios` (needs riscv-gcc). The host variant (harness_csr at base 0, no pumice APB cfg) is still TODO.

9 Measured Results

Every number in this chapter is from the Nexys A7, not simulation: 75 MHz controller clock, DDR2-300, BL4 on the x16 MT47H64M16, `open_page` policy, the 4+4 generator bitstream. **Peak is 600 MB/s** — 8 bytes per controller cycle. Measured 2026-09-14; sections 8.2 and 8.5 re-measured 2026-09-26 on the post-RBL-removal bitstream; the page-friendly and page-thrash rows re-verified 2026-10-03 on the post-sdpram-fix image (`char_postfix_2026-10-03.csv` — the fix is on the harness UART/debug SRAM, off this datapath: row_major bl16 read 572.3 MB/s, close_page `col_major` bl8 read 195.2 MB/s to the digit of the 2026-09-26 campaign). The AxLEN/outstanding sweeps of 8.1/8.2 were not re-run post-fix; treat them as pre-fix dated.

The figures and the raw records are in the repository, so nothing here has to be taken on trust: `build-perf/reports/bank_gap_sweep.json` (192 records), `axlen_sweep.json`, `outstanding_sweep.json`, and the plots under `build-perf/reports/plots/`.

9.1 8.1 Streaming bandwidth, and why short bursts fall short

Reads and writes both reach about 95% of peak once bursts are long enough. They do not at short burst lengths, and the reason is **latency, not overhead**.

Read bandwidth against AXI burst length, at 8 bursts outstanding

AxLEN	read MB/s	% peak	read latency (cycles)	Little's-law prediction
1	98.7	16.4%	50.6	93.1
2	196.6	32.8%	48.0	192.0
4	367.9	61.3%	50.0	355.5
8	575.0	95.8%	49.8	570.0
16	575.8	96.0%	96.0	570.0

The model is `min(outstanding x AxLEN / (latency + AxLEN), 0.95) x peak`. With 8 bursts in flight against a ~49-cycle read, a one-beat burst simply cannot fill the pipe — the requester runs out of things to have asked for.

That is an inference from one curve, so it was tested directly from the other axis: hold AxLEN and sweep the number of bursts in flight instead.

The short-burst shortfall recovers completely when more reads are in flight

AxLEN	at 8 outstanding	best reached	at
1	98.7 MB/s (16.4%)	390.6 MB/s (65.1%)	32, still climbing

AxLEN	at 8 outstanding	best reached	at
2	196.6 MB/s (32.8%)	**573.1 MB/s (95.5%)**	24
4	367.9 MB/s (61.3%)	**574.6 MB/s (95.8%)**	16
8	575.0 MB/s (95.8%)	575.6 MB/s (95.9%)	12

This is the decisive result. AxLEN 2 and 4 reach the *same* ceiling as AxLEN 8 once enough reads are in flight. A per-transaction penalty could not behave that way — it would scale with the number of transactions and never go away. So nothing in the controller's datapath limits short bursts; the read latency and the requester's budget are the whole story. AxLEN 1 needs roughly 49 bursts in flight and the harness ceiling is 32, which is why it alone is still climbing rather than a different mechanism.

The knee confirms the shape too. It is the smallest number of bursts in flight at which bandwidth reaches 95% of the plateau, and the model puts it at `0.95 x (latency + AxLEN) / AxLEN`:

Where bandwidth reaches its plateau, measured against the model

AxLEN	model knee	measured knee	ratio
1	47.7	none inside 32 (still climbing)	—
2	24.1	24	0.99x
4	12.7	12	0.95x
8	6.9	8	1.17x

Scaling and constant both hold. The 1.17x at AxLEN 8 is the sweep grid rather than a discrepancy — the model wants 6.9 and the available steps are 4 and 8. The per-point agreement is as good: at AxLEN 4 the model tracks all eight points from -0.1% to +4.4%, with measured bandwidth sitting slightly *above* the prediction throughout, which says the effective latency is marginally better than the sampled average rather than worse.

> **A note on how this knee is scored, because the obvious way is wrong.** Reading the knee as "the first point that stopped improving" names the point *after* saturation — one sweep step late, every time. Scored that way these same measurements read 32 / 24 / 12 and appear to sit 1.3-2x above the model, which is exactly what an earlier draft of this chapter reported as an open anomaly. There was no anomaly. The knee is where bandwidth *arrives* at the plateau, not where it first fails to climb.

9.2 8.2 How much idle the controller absorbs

The gap sweep runs N writers and N readers **concurrently**, one bank apiece, inserting 0 to 15 idle clocks between bursts, across three address orders and at 4+4 / 3+3 / 2+2 / 1+1 engines. It answers a different question from the bandwidth sweeps: not “how fast can it go” but “how much slack does it have”.

The knee is the largest gap whose read bandwidth is still within 3% of the gap-0 value.

Gap knee by address order and generator count (re-measured 2026-09-26)

address order	4+4	3+3	2+2	1+1
cacheline	15	15	10	1
row-major	15	15	10	0
col-major	15	15	10	1

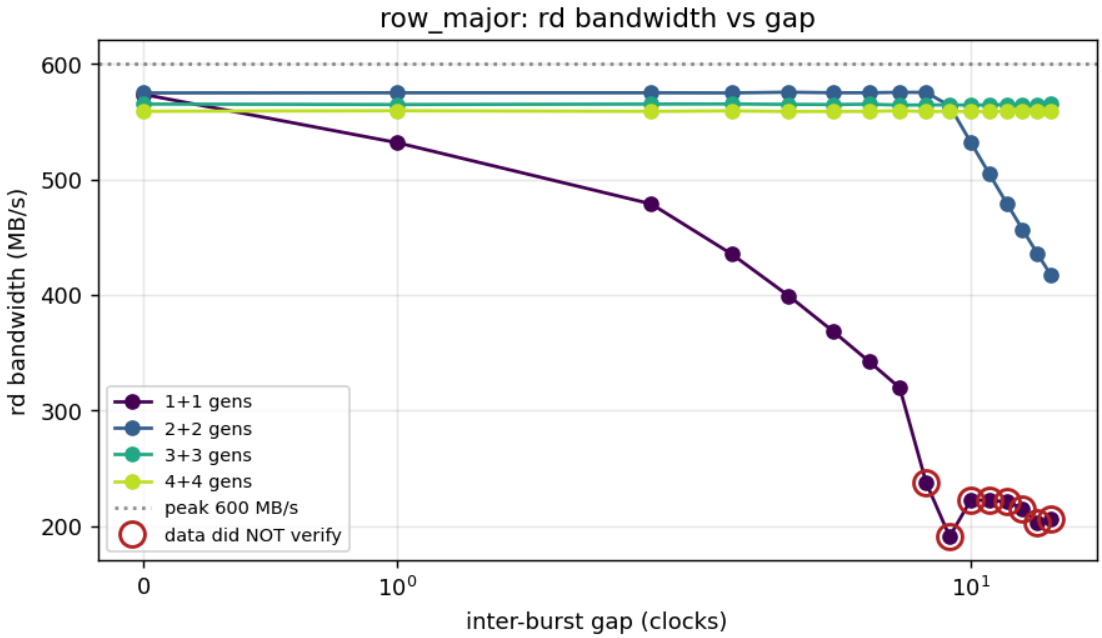
Bus bandwidth from gap 0 to gap 15, all three address orders agreeing within 2 MB/s: 1+1 570.5 -> 237.2 (42% retained), 2+2 564.8 -> 442.7 (78%), 3+3 560.7 -> 560.8 (100%), 4+4 549.9 -> 549.8 (100%).

Two corrections to the 2026-09-14 version of this table.

col-major is no longer flat. It previously read 15 at every generator count — insensitive to injected idle — and now tracks the other two orders exactly. Nothing about the gap changed; col-major got FASTER. It used to be DRAM-bound, slow enough that idle cycles inserted between bursts were absorbed by work the controller was already behind on. The read-ceiling fix and the bank-timer lookahead removed that slack, so col-major now runs close enough to the bus limit to feel a gap like anything else. A knee that *appears* when a controller improves is the normal direction of travel, not a regression.

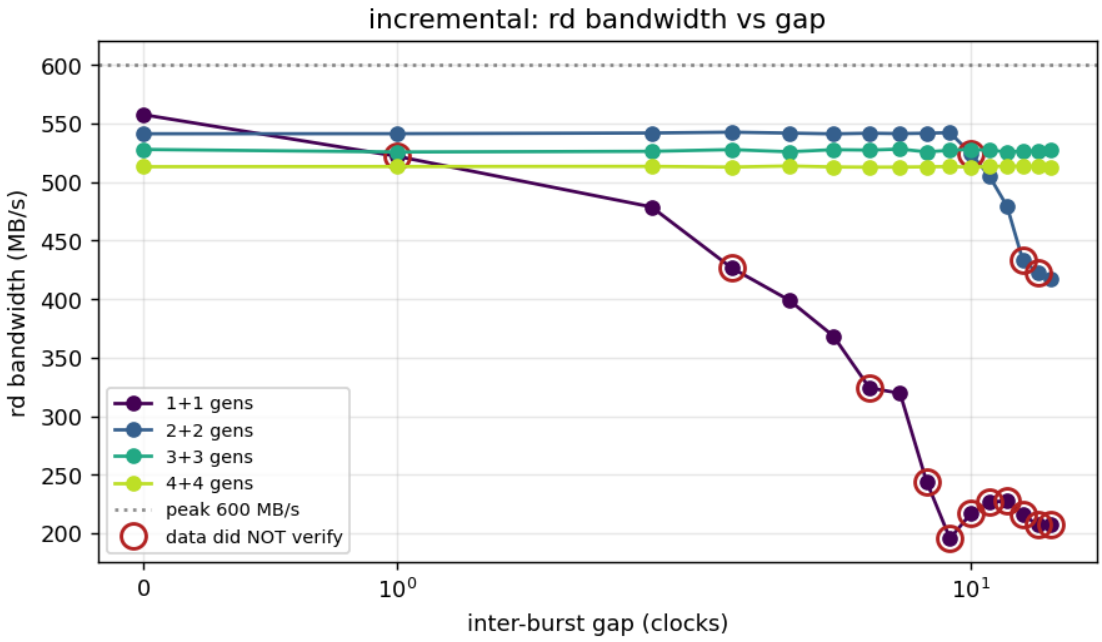
The figures quoted below the table were labelled “read bandwidth” and are BUS bandwidth — both directions’ bytes over the measurement window. The distinction matters because `rd` and `wr` in the JSON are each measured over their OWN active cycles, so `bus` is NOT `rd + wr`: at 1+1 gap 15 the read engine finishes in 46,034 cycles while the write engine needs 80,989, and subtracting the two per-direction figures to “recover” the other invents an asymmetry that is not there.

9.2.1 Figure 8.1: Read Bandwidth vs Gap, Row-Major



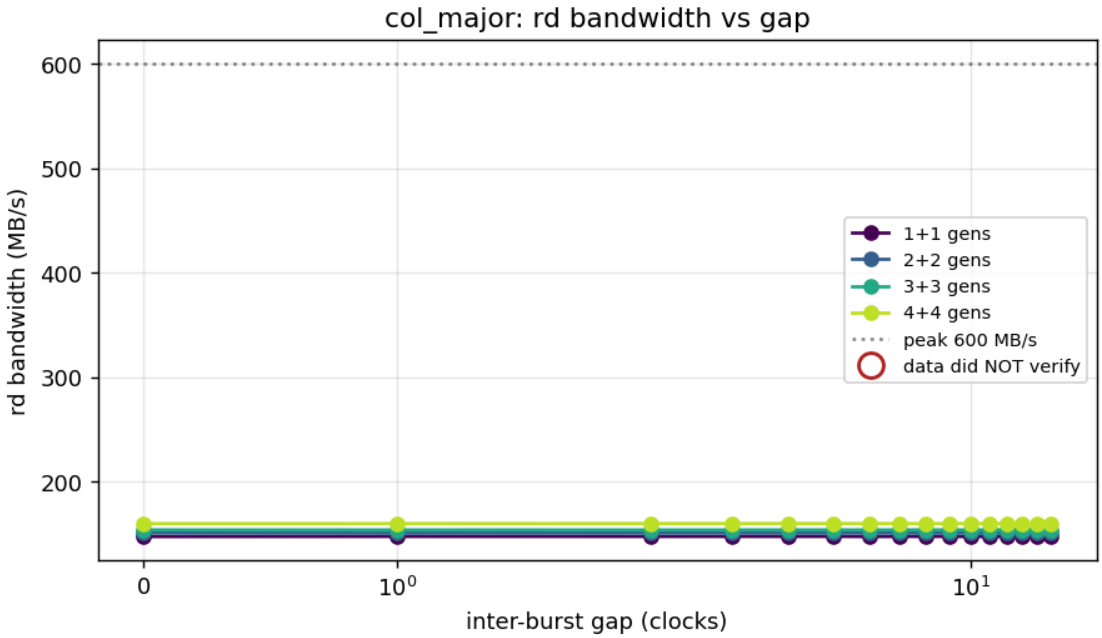
Read bandwidth against inter-burst gap, row-major, one line per generator count

9.2.2 Figure 8.2: Read Bandwidth vs Gap, Cacheline



Read bandwidth against inter-burst gap, cacheline order

9.2.3 Figure 8.3: Read Bandwidth vs Gap, Col-Major



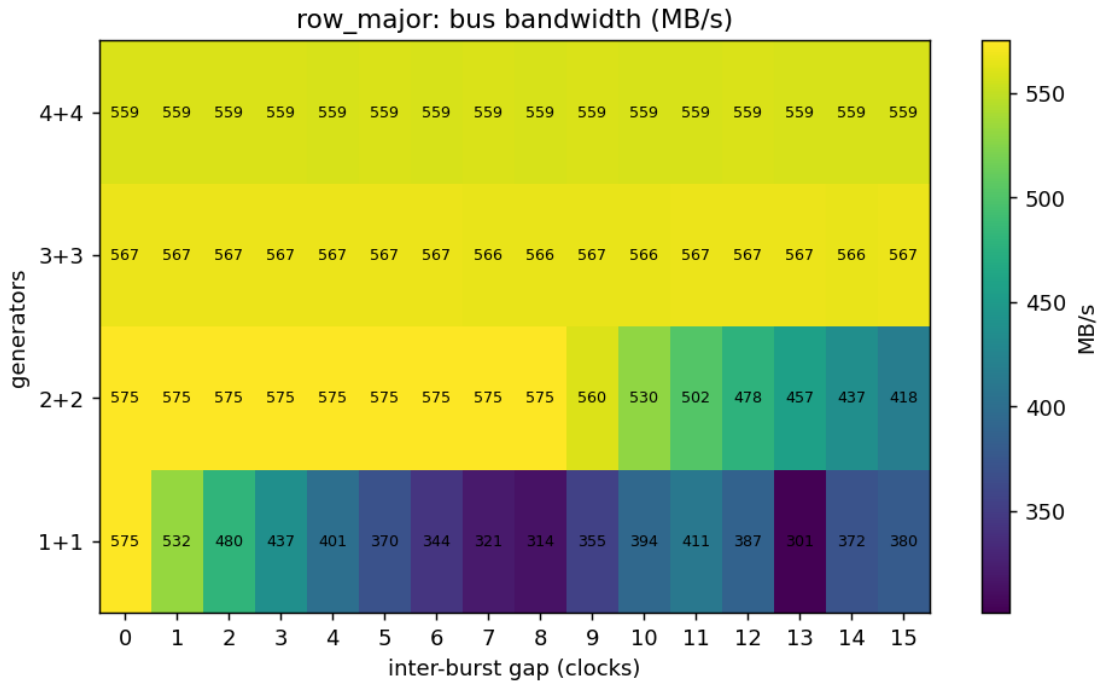
Read bandwidth against inter-burst gap, col-major order

Reading these curves. Two things look wrong at first glance and are not:

The high generator counts are flat. A gap only bends the curve once aggregate demand falls **below** what the controller can deliver. At 4+4, gap 15 still leaves each generator at roughly a 52% duty cycle, and four of those together still out-demand the controller — so nothing moves. A flat curve there is a result, not a broken axis. It also means the 4-bit gap field cannot inject enough idle to starve four generators: at high counts the lever is the outstanding dial, not the gap.

Col-major is flat at every generator count. It is page-miss bound at about 150 MB/s, 25% of peak. The DRAM is the limit, so generator pacing never becomes the binding constraint. Compare it against row-major's 575 MB/s on the same silicon and the same settings: **a 3.8x spread from the address pattern alone**, which remains the single most useful fact in this guide. Tune the address map before tuning anything else.

9.2.4 Figure 8.4: Bus Bandwidth Across Gap and Generator Count



Heatmap of total bus bandwidth against gap and generator count, row-major

---

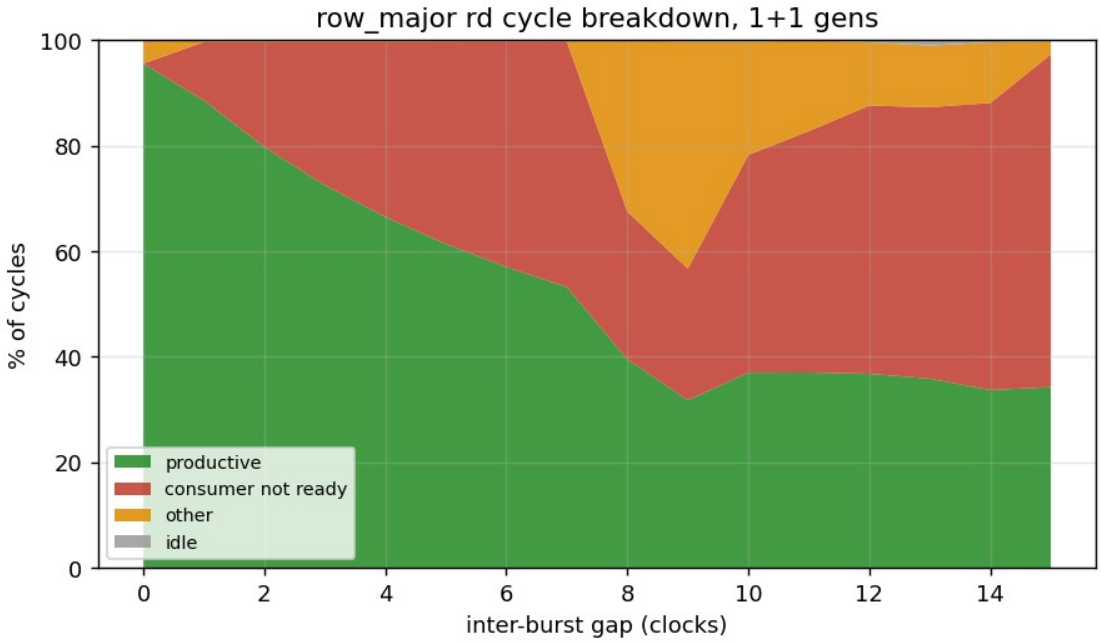
9.3 8.3 Where the cycles went

Bandwidth says a point was slow. The cycle classification says *who stopped*, which is the difference between a finding and a number.

Read cycle breakdown, row-major, 1+1 generators

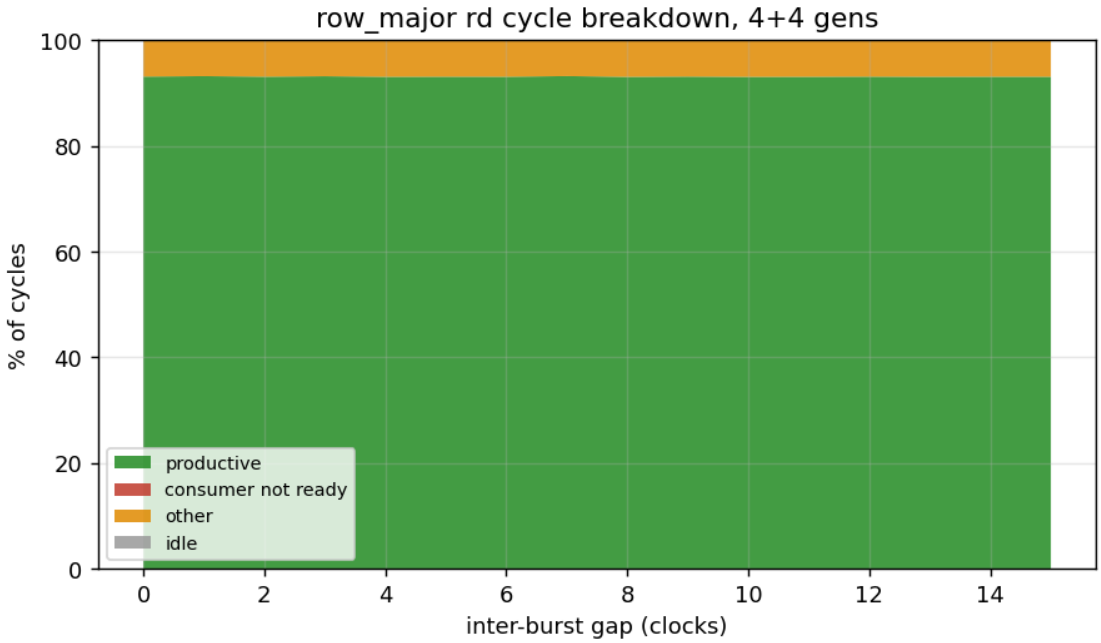
gap	read MB/s	productive	consumer not ready	other
0	574	95.6%	0.0%	4.4%
4	399	66.5%	33.2%	0.2%
8	237	39.5%	28.0%	32.5%
15	206	34.3%	63.1%	2.6%

9.3.1 Figure 8.5: Cycle Breakdown at 1+1 — the Bend and its Cause



Read cycle breakdown against gap at one generator per direction

9.3.2 Figure 8.6: Cycle Breakdown at 4+4 — Saturated at Every Gap



Read cycle breakdown against gap at four generators per direction

At 1+1 the productive fraction falls from 95.6% to 34.3% while “consumer not ready” climbs from nothing to 63.1%. That is the engine’s own inter-burst gap deasserting `rready`, so **the bandwidth went to generator-induced idle, not to anything the controller did**. At 4+4 the productive fraction sits at 93.1% regardless of gap — the controller is saturated everywhere on that axis, which is the same fact Figure 8.1 shows as a flat line, seen from the cause side.

> **Two traps in this instrument, both of which produced published nonsense before they were caught.**
>
> The bus meter **free-runs** from `clear_stats()` until it is read, so its own total spans the host’s UART chatter as well as the transfer — 7.2M cycles against a 16.7k-cycle window on a 1+1 point, a factor of 430. Fractions taken against that total made a point moving 95% of peak report “0.2% productive, 99.8% starvation”. The *counts* are sound — productive lands on exactly the beats moved — so the fix is the denominator: normalise against the timer’s first-to-last stamp for that direction. The meter’s own starvation bucket is where the host idle lands and is not reported as a fraction at all.
>
> On a **read**, the engine is the *consumer*, so its own pacing gap shows up as `rvalid && !rready` — the bucket a write would call backpressure. It is generator-induced idle, the opposite of the controller refusing. The label in Figure 8.5 says “consumer not ready” for exactly this reason; do not read the read and write breakdowns with the same vocabulary.

9.4 8.4 Integrity, and one configuration that must not be quoted

All 192 gap-sweep points completed. **22 of them returned mismatching data**, and they are not scattered: they are a specific configuration, tracked as pumice BUG-014 (was PUMICE-037).

Concurrent read+write with the **reader’s** gap at 8 or above returns wrong data, and when the two address ranges overlap it leaves genuinely corrupted cells behind. The boundary was narrowed by elimination on the board, each line its own run from a verified-clean 128 MiB prefill:

pumice BUG-014 (was PUMICE-037) — narrowing the failing configuration

configuration	result
prefill, then read every bank read-only, all three orders	clean
reader alone, gap 0-15	clean at every gap
writer alone, gap 0-15	cells clean

configuration	result
writer gap 0-15, reader gap 0	clean at every writer gap
reader gap **0-7** with a concurrent writer	clean
reader gap >= **8** with a concurrent writer	1.5k-7k of 32000 beats wrong
4+4 concurrent at gap 0 and gap 4	clean, cells clean

Neither engine alone does it at any gap, and the writer's own gap never does it. **Everything in this chapter is at gap 0-7 and is unaffected**, but no gap >= 8 concurrent point may be quoted as a clean operating point. The plots ring those points, and `plot_bank_gap.py` refuses any record file whose bandwidth exceeds its own stored ceiling.

Whether the defect is in the controller or in the harness read engine is not yet settled. The stray-beat counter is zero on all 22 failing points, which rules out the read engine's stray-drain path and points at the data itself; against that, a read engine cannot corrupt cells, and cells are corrupted. The next step is a simulation reproduction with waves — and the reason this reached the board at all is that both gap-bearing simulation suites drain the writer before starting the reader, so **concurrency with a gap has never been simulated**.

9.5 8.5 What the three runtime modes are worth (2026-09-26)

Chapters 8.1-8.4 measure the controller at its default settings. This section measures the **runtime surface itself** — the three orthogonal axes that exist so a scheduling policy can be switched on and compared on real hardware. 169 points, txn_scale=1000, all integrity-clean.

The headline is a negative result, and it is worth more than a positive one would be: **reordering is the only thing in this controller that pays. Everything layered on top of it is inert or harmful.**

9.5.1 Axis 1 — scheduling order

Measured at **equal page policy**, which had never been done. The stock `inorder` preset also pins CLOSE page, so the axis used to read as a ~16x deficit that was mostly the page policy rather than the ordering.

Read MB/s against 600 MB/s peak, order_mode at equal page policy

config	cacheline	col-major
`open_page` (FR-FCFS)	**561.3**	**195.2**
`inorder_open`	143.0	102.4

FR-FCFS is worth 3.9x on streaming and 1.9x page-hostile. That is the reorder CAM earning its area, and it is the largest single effect in the design.

The axis-1 *sub*-policies are a different matter. Measured single-direction so that read/write turnaround is not the limiter, **eight of ten land within ±0.3% of the default**. `row_most_pending` is the only lever with a real effect and it is a net loss: +1.1% on sequential traffic, **−19.4%** on page-hostile, +7,300 activates, and 2.4x read latency in the concurrent run. It selects the bank with the most queued work rather than the one whose row is already open, so it works against the page policy. The default oldest-first wins on both families.

> **Why the first attempt at this measured nothing.** Run concurrently at 4w+4r, all ten sub-policies returned identical bandwidth. The stall counters showed why: `limiter = turnaround` at 84% blocked. An arbiter sub-policy chooses WHICH COMMAND, not which direction, so no tie-break can move a workload bound by a global DQ constraint. The sweep was flat for a reason that had nothing to do with the knobs. **A policy sweep on a workload that cannot express the policy measures nothing, and looks exactly like a policy that does nothing.**

9.5.2 Axis 2 — page policy

Every predictor lands within ±0.1% of plain open page, on all four address families, sequential and concurrent.

Modes 6/7 (`rbl_static` / `rbl_dyn`) were **retired 2026-09-26** after being measured on a workload built specifically to suit them: three generators confined inside a row against one striding across rows, all on the same bank, which is the per-row locality variation a per-row predictor needs and which no uniform pattern provides. The mechanism **worked** — thrash fell 100% -> 57.8%, conflict-activates becoming empty-activates — and it still lost, because mode 6 paid +22,827 activates to save precharges that never materialised (−26% bandwidth) and mode 7's hill-climb drove its threshold to "never close early", landing bit-identical to plain open page.

Modes 4/5 (`adapt_time`, `adapt_access`) are **kept but unproven**. They show no benefit, but that is not the same finding: their triggers may never have fired. `adapt_time` closes on an idle timeout and every workload here saturates the generators; `adapt_access` needs its 2-bit counters to learn. Retiring them on absence-of-benefit would repeat the error that made RBL unmeasurable for months.

9.5.3 Axis 3 — refresh

Measured on **open** page. The stock `refresh` profile pins close page (~46 MB/s), which measures refresh in the regime where it matters least.

Read MB/s against 600 MB/s peak

config	row-major	vs default
default tREFI	572.0	—
`refresh_credit_open`	575.3	+0.6%
`fast_refresh_open`	536.6	**−6.2%**
`slow_refresh_open`	**599.0**	**+4.7%**

Refresh costs 4.7% of streaming bandwidth. Relaxing tREFI recovers it and reaches **599.0 MB/s — 99.8% of the 600 MB/s ceiling**, the highest number this controller has produced on hardware. Refresh credit is inert.

This is the one tunable on any axis that pays, and it is a JEDEC timing parameter rather than a scheduling policy.

9.5.4 Multi-ID traffic

`col_major_bl8_multiid` (LFSR-driven IDs) is **bit-identical to single-ID** on every controller config. FR-FCFS already reorders across the whole CAM regardless of ID, so ID diversity exposes no additional opportunity.

9.6 8.6 Reproducing these numbers

```
cd build-perf
make bitstream && make program
python3 bin/axlen_sweep.py                  # 8.1, first table
python3 bin/outstanding_sweep.py            # 8.1, second table
python3 bin/bank_gap_sweep.py               # 8.2-8.4, ~12 minutes
python3 bin/plot_bank_gap.py reports/bank_gap_sweep.json

# 8.5, the runtime axes -- pass the clock EXPLICITLY, see the warning
below
cd host
for p in order sched_sub paging pairs_refresh_open matrix; do
  python3 pumice_master.py --char --char-scale 1000 --char-profile $p
\
      --no-level --clk-mhz 75 --csv ../reports/axis_$p.csv
done
```

> **`--clk-mhz` defaults to 66.667 and this build runs at 75.** Bandwidth is computed as `(bytes / cycles) * clk_mhz`, so the default silently scales every number by 0.889 and mis-states the ceiling as 533 MB/s. Pass it, or

let `pumice_char.resolve_clk_mhz()` read `clk_hz` out of the bitstream. The same class of error in the other direction once reported open-page reads at 123% of what the port can physically carry.

Read section 8.5 with the `limiter` column, not only the bandwidth column. A policy sweep on a workload that cannot express the policy returns a flat table that is indistinguishable from a policy with no effect.

Leave `PREFILL` at its default (`point`). It re-fills the whole device before every measurement so that points are independent; without it, a point that corrupts cells is inherited by every later point that reads the region and reported as *that* point's result. Chapter 3 lists the rest of the pitfalls, and `docs/DDR2_BANDWIDTH_MEASUREMENT.md` carries the full treatment.